Energy transport: established convention and open balances

Classical local thermoelasticity · outward-positive flux

$$\partial_t \mathcal{E} + \nabla\cdot \mathbf{J} = \rho\, \mathbf{f}\cdot\mathbf{v}$$

$$\mathcal{E} = \tfrac{1}{2}\rho|\mathbf{v}|^2 + \rho e \quad J_i = q_i - \sigma_{ij}v_j$$

Derived from momentum balance + first law; e is specific internal energy. No internal heat source shown.

Multiphysics field contributions

Electrical / magnetic power

Thermal-rate and memory storage

Kernel- and domain-dependent boundary power

OPEN: derive for each stated constitutive model.

Interface energy accounting

Use the complete, theory-specific flux.

State boundary passivity and storage.

Evaluate the total field, including cross terms.

No universal modal identity is asserted here.

This schematic replaces unverified generalized flux and evanescent-mode claims in the earlier artwork.